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CARDON et al.(10) **Pub. No.: US 2025/0285256 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **APPARATUSES AND METHODS FOR
LARGE AREA WIRELESS FUSELAGE DENT
INSPECTION****Publication Classification**

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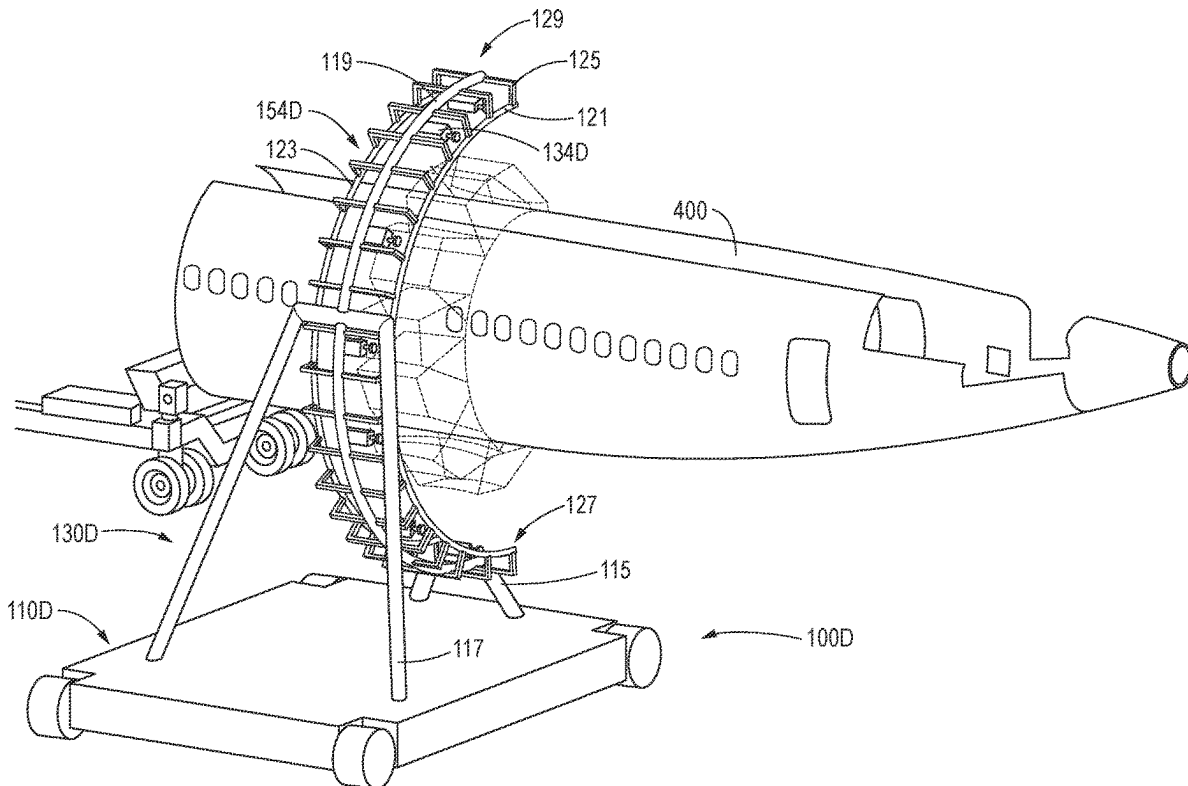
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ABSTRACT

The present disclosure provides an inspection cart for inspecting large area objects, such as airplane fuselages. In one aspect, an inspection cart includes a tower supporting a track. The inspection cart also includes a carriage movable along the track. The carriage supports a 3D scanner having at least two cameras and a projector. The inspection cart also includes integrated optical fiducials arranged relative to the track so that the optical fiducials are in a field of view of the 3D scanner during scanning of an object. In other aspects, an inspection system that includes at least one inspection cart is provided. In further aspects, a method of inspecting a large area object using an inspection cart is provided.

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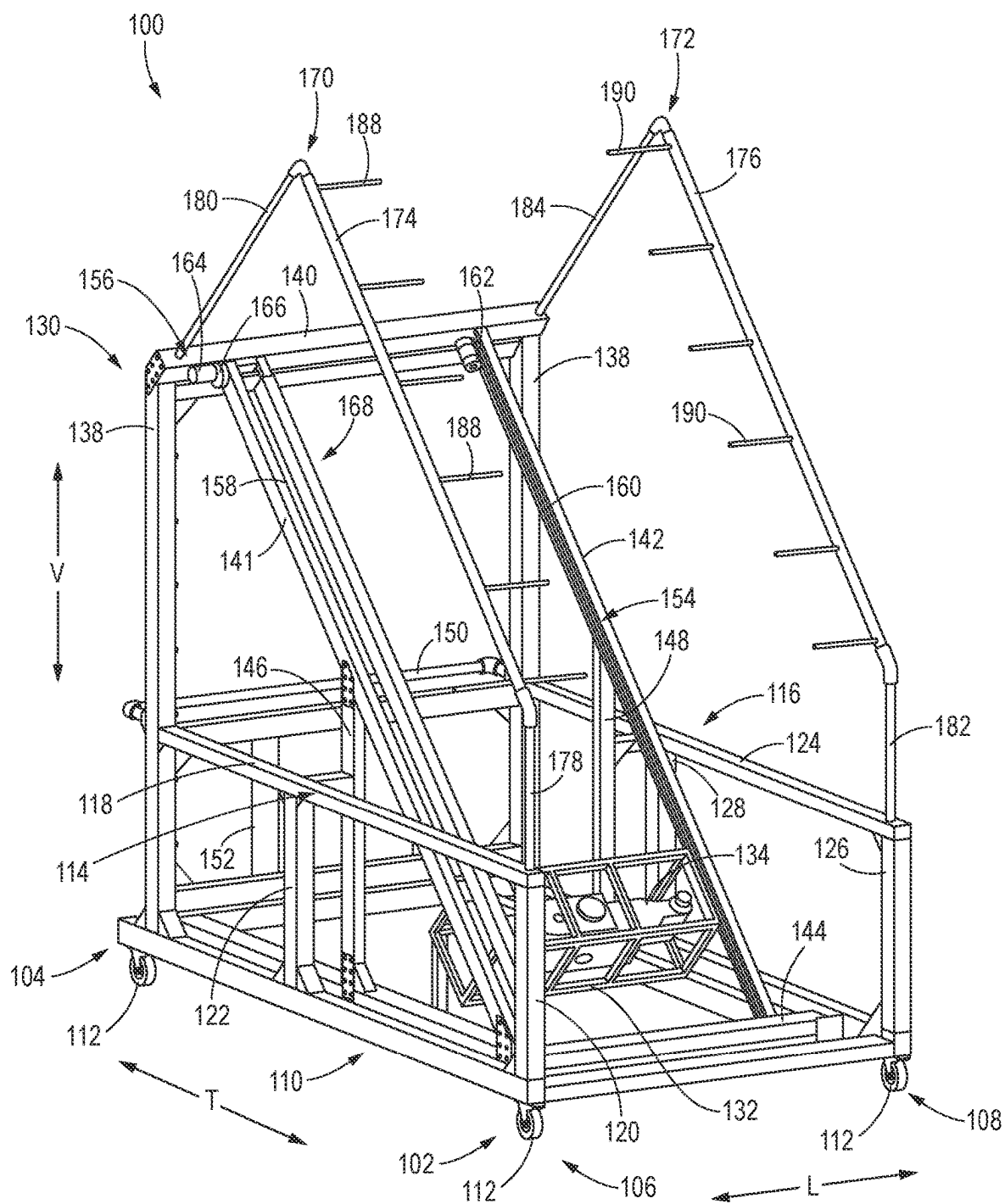


FIG. 1

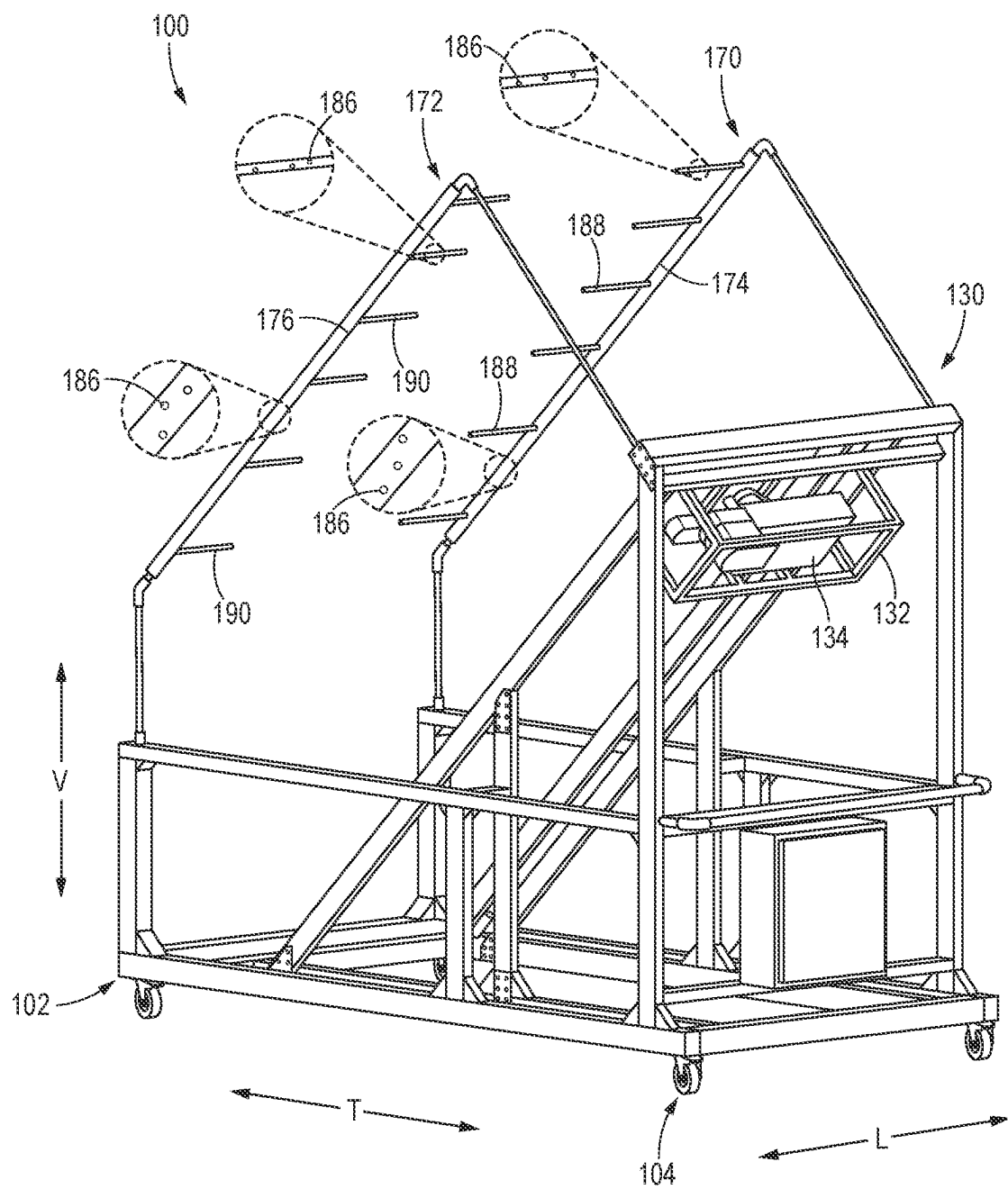


FIG. 2

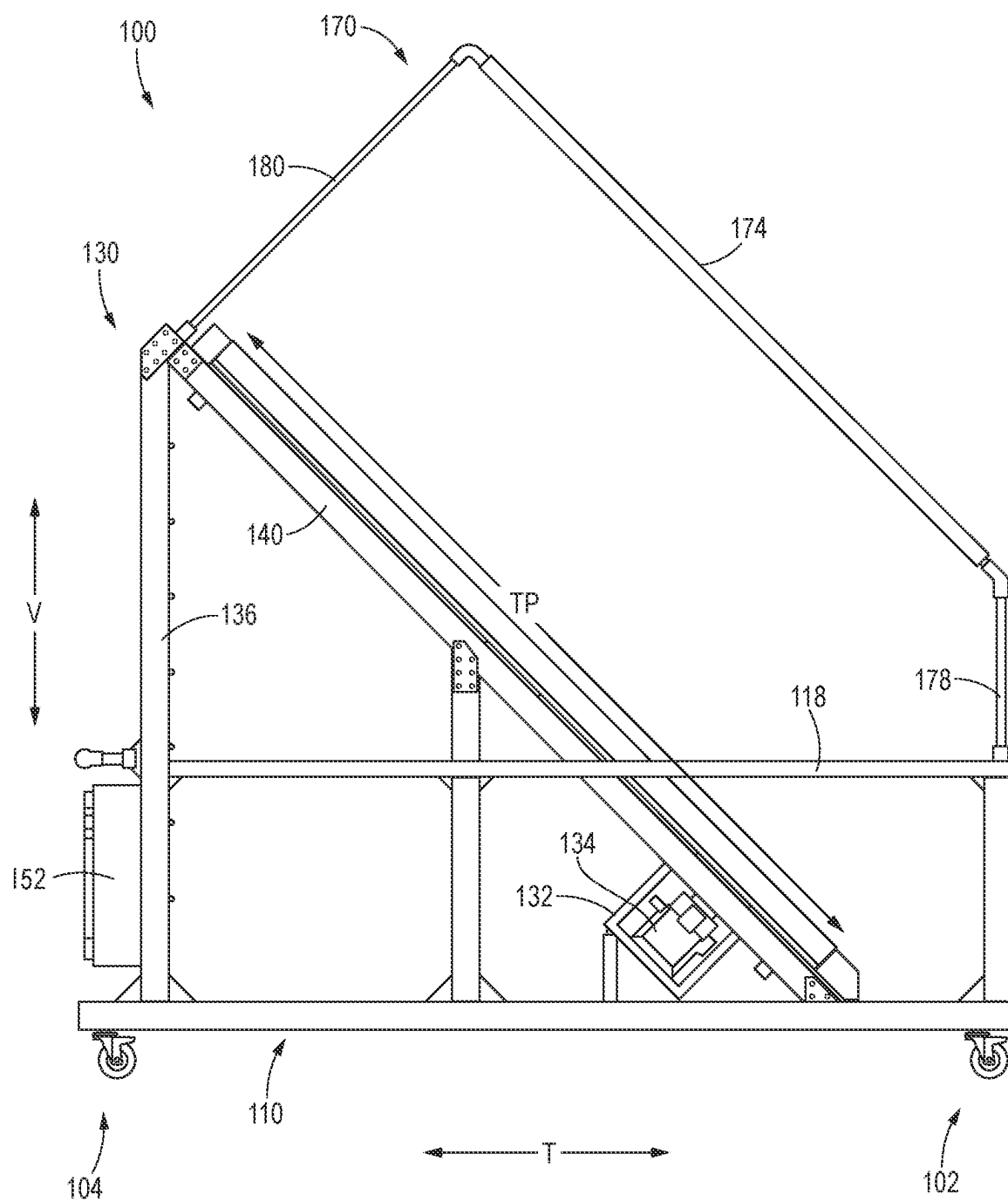


FIG. 3

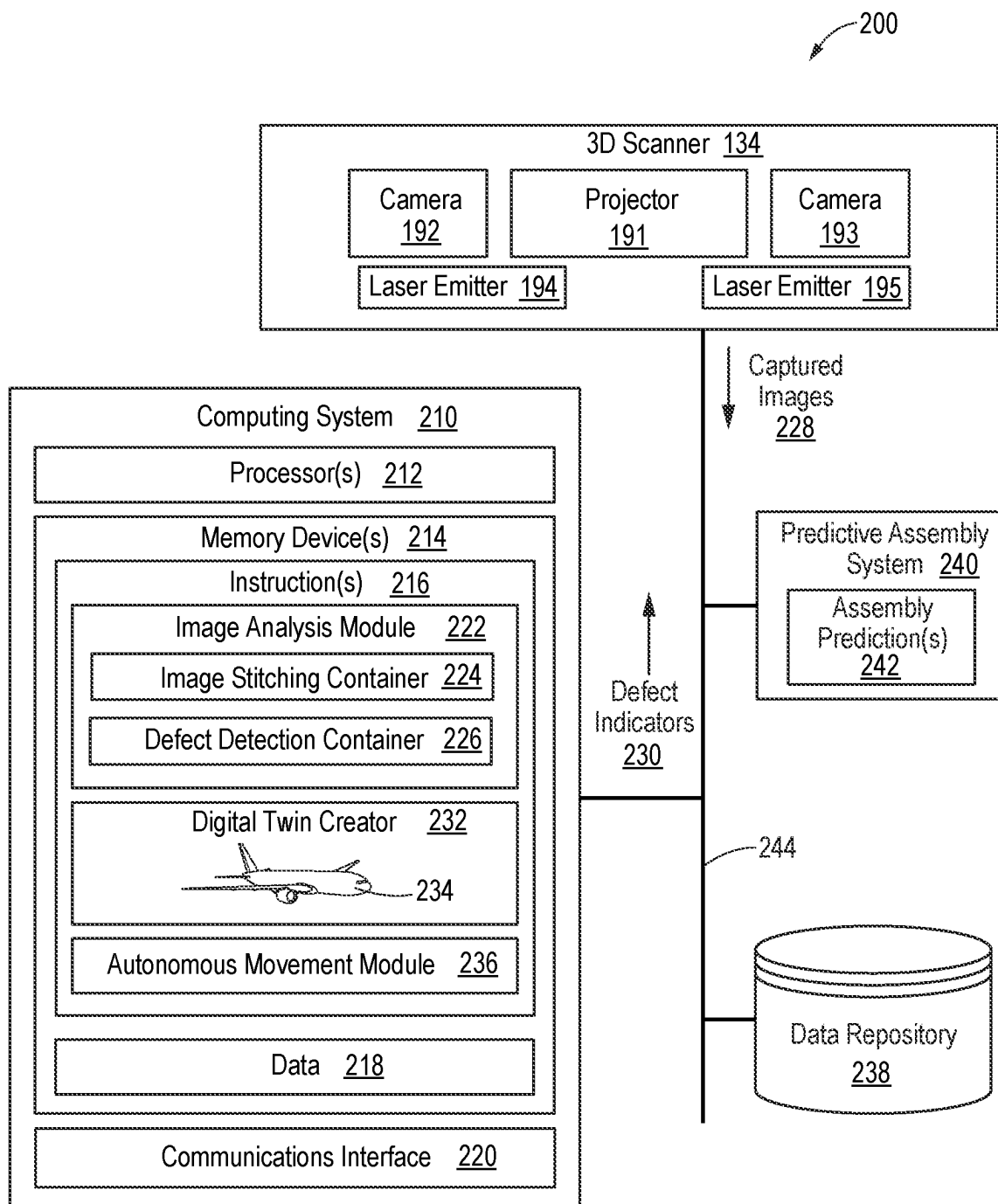


FIG. 4

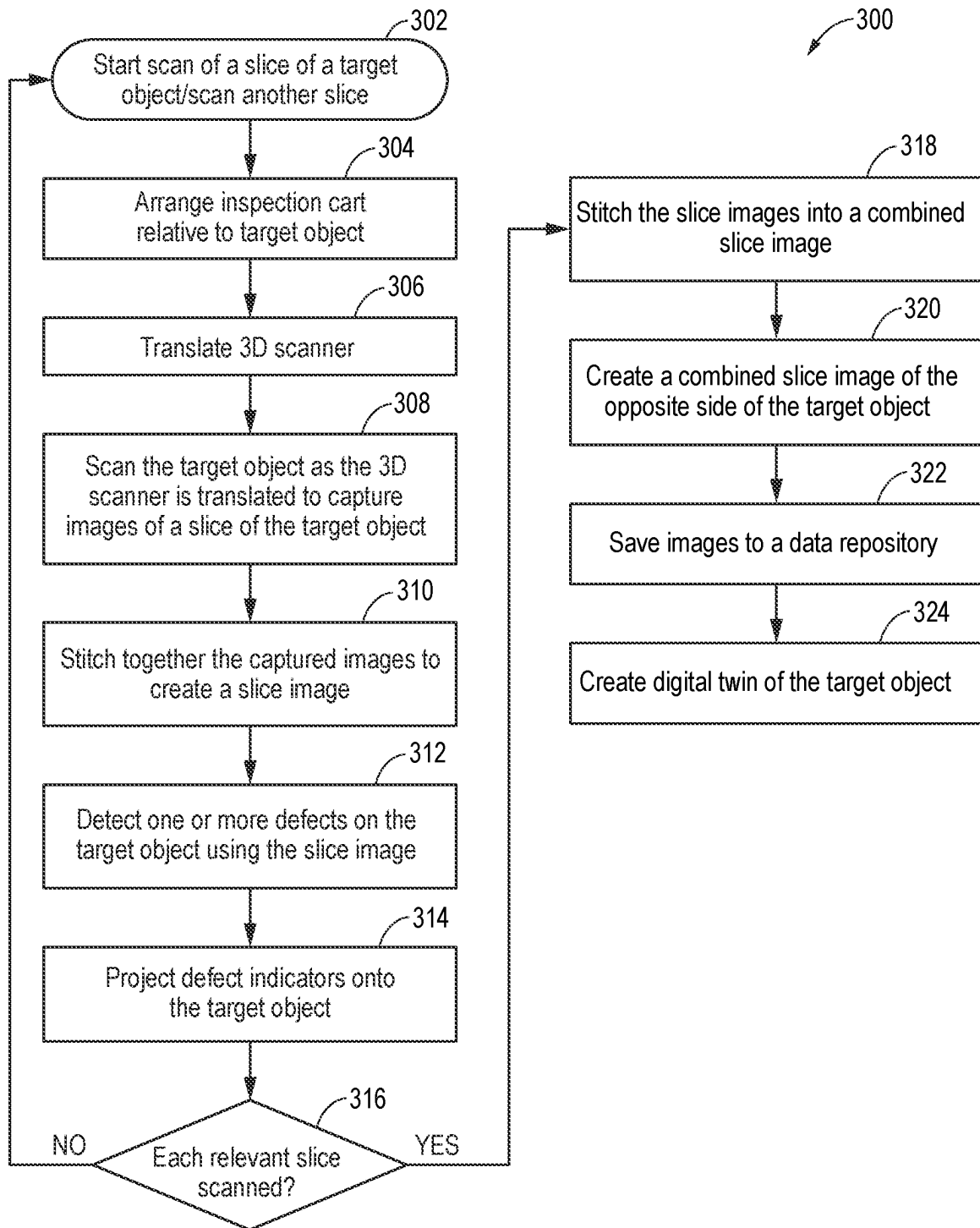


FIG. 5

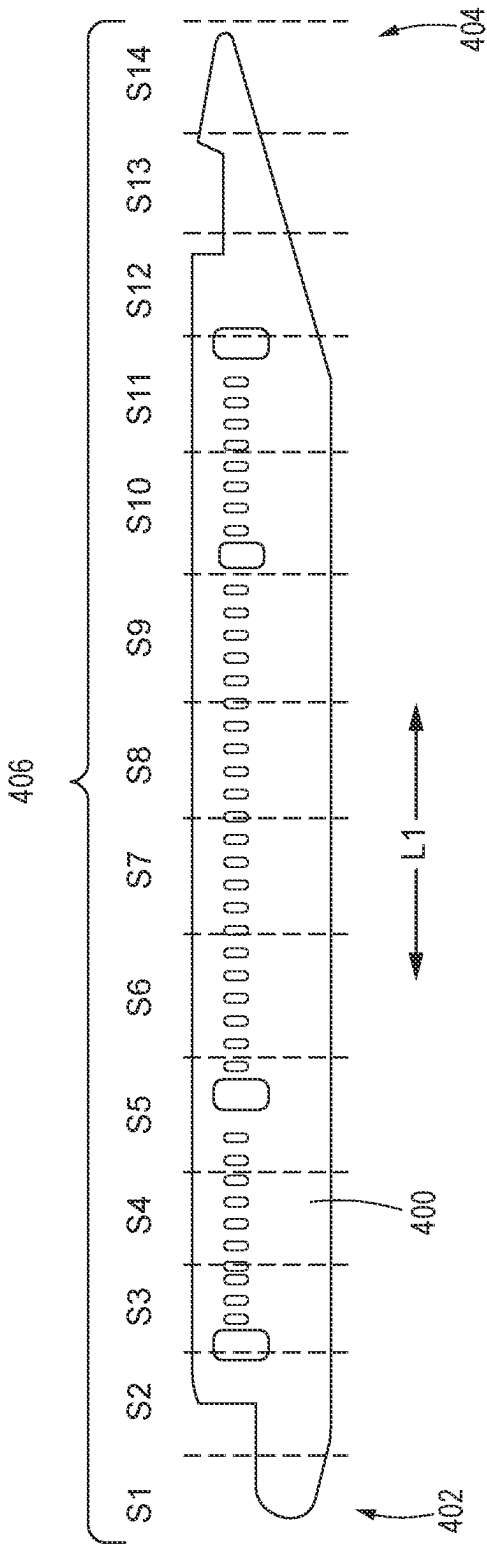


FIG. 6

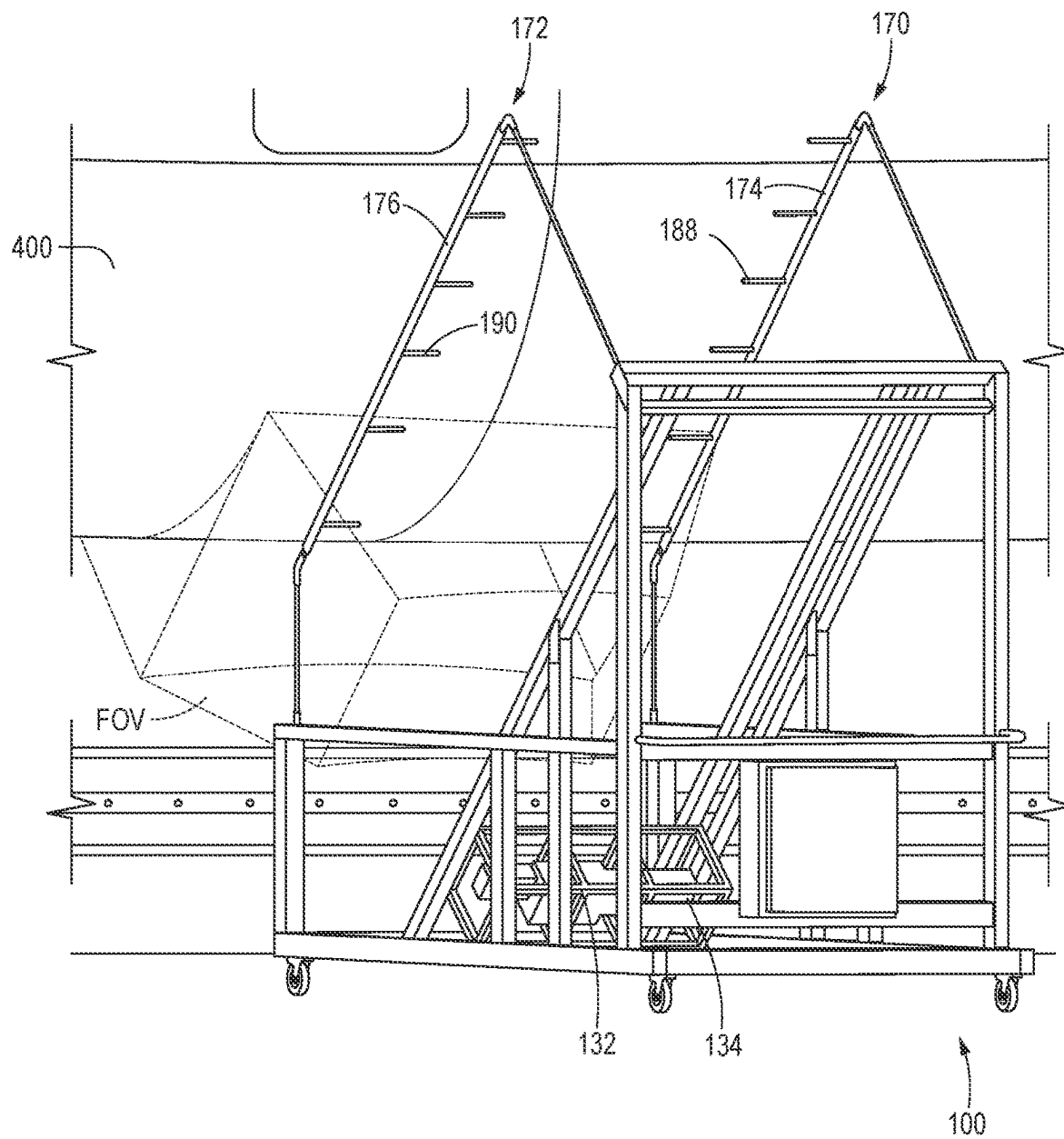


FIG. 7

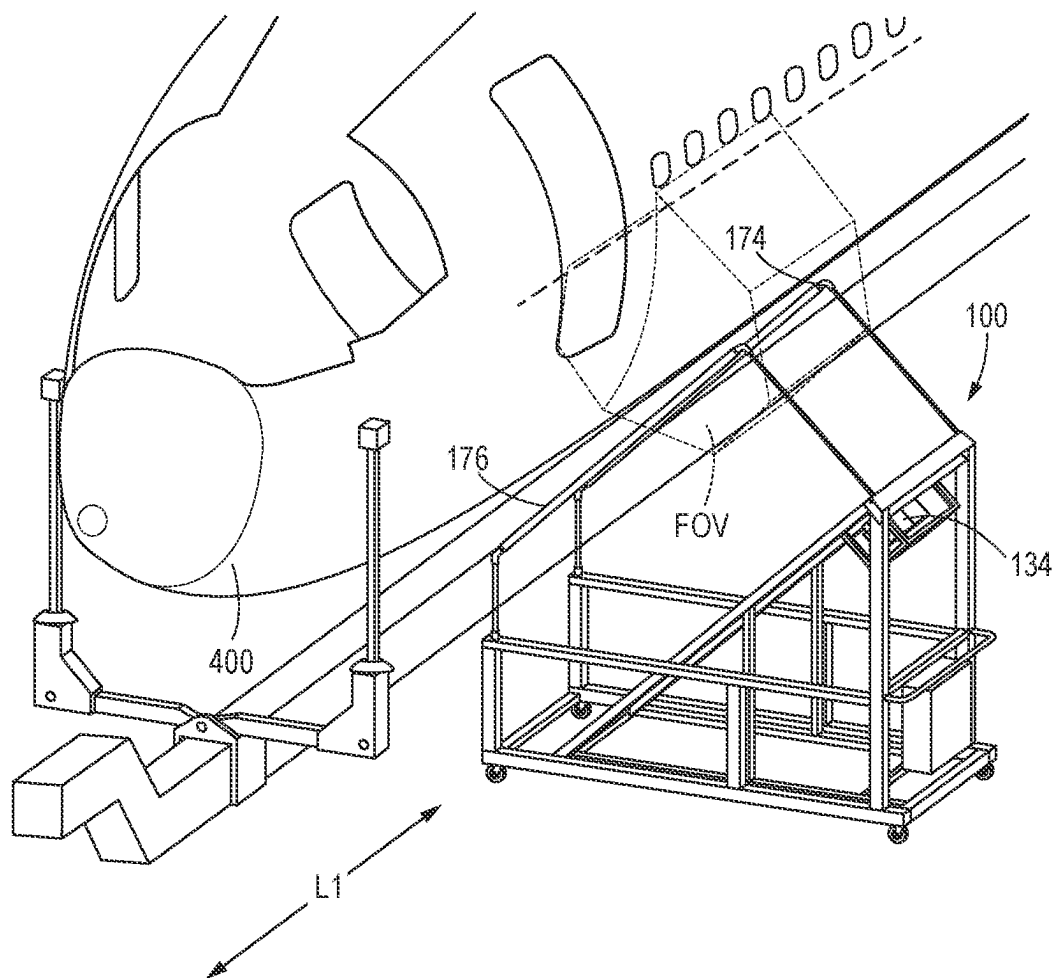


FIG. 8

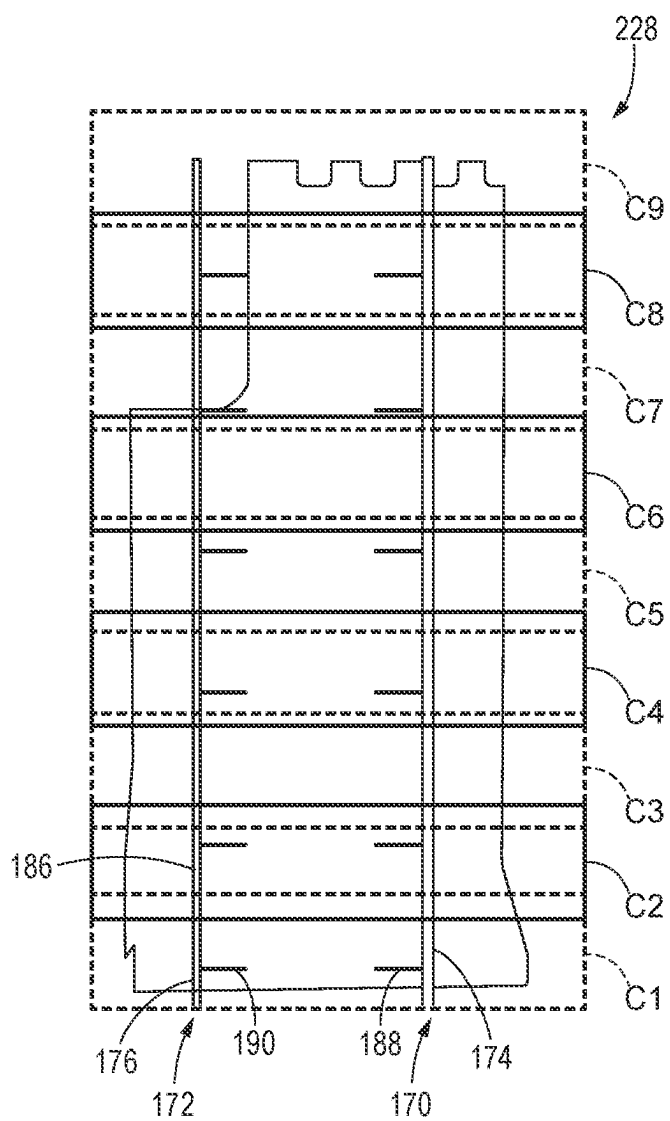


FIG. 9

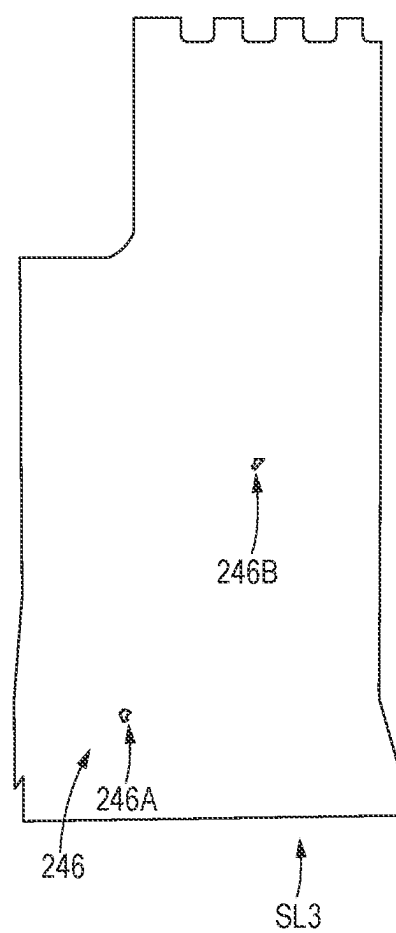


FIG. 10

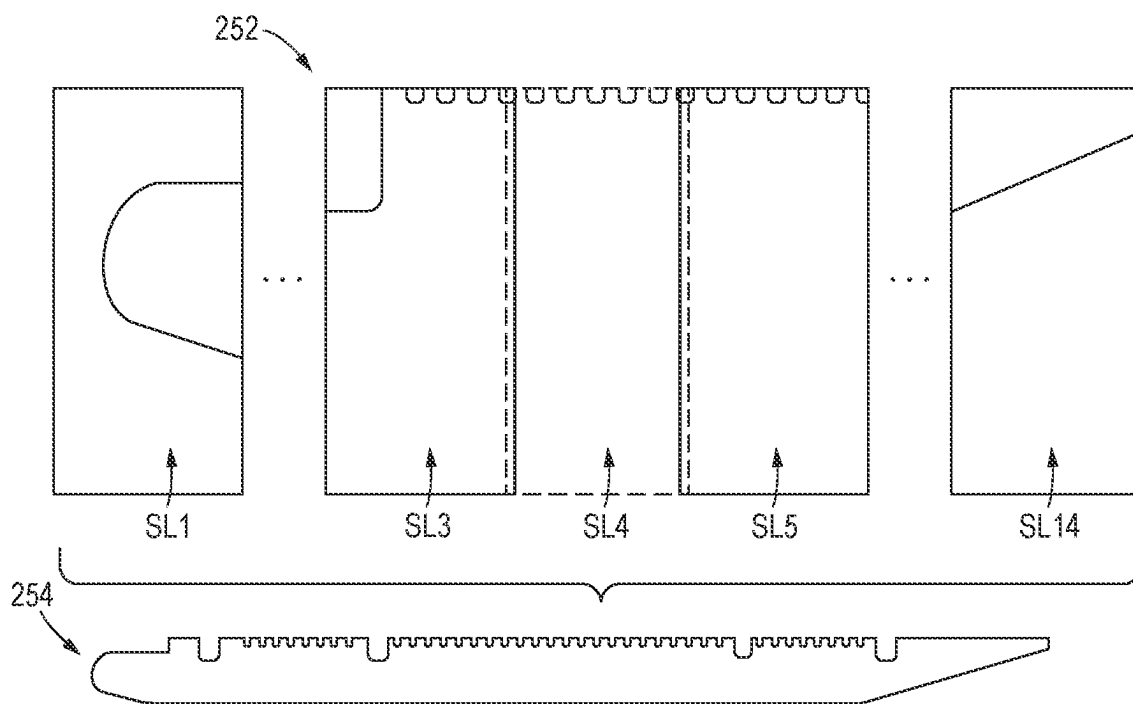
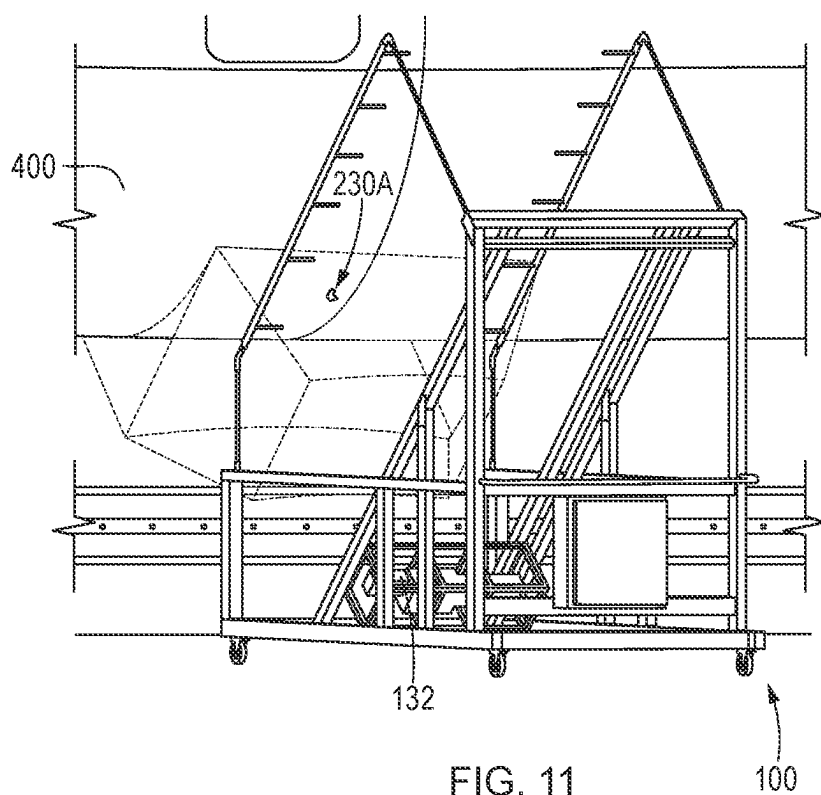


FIG. 12

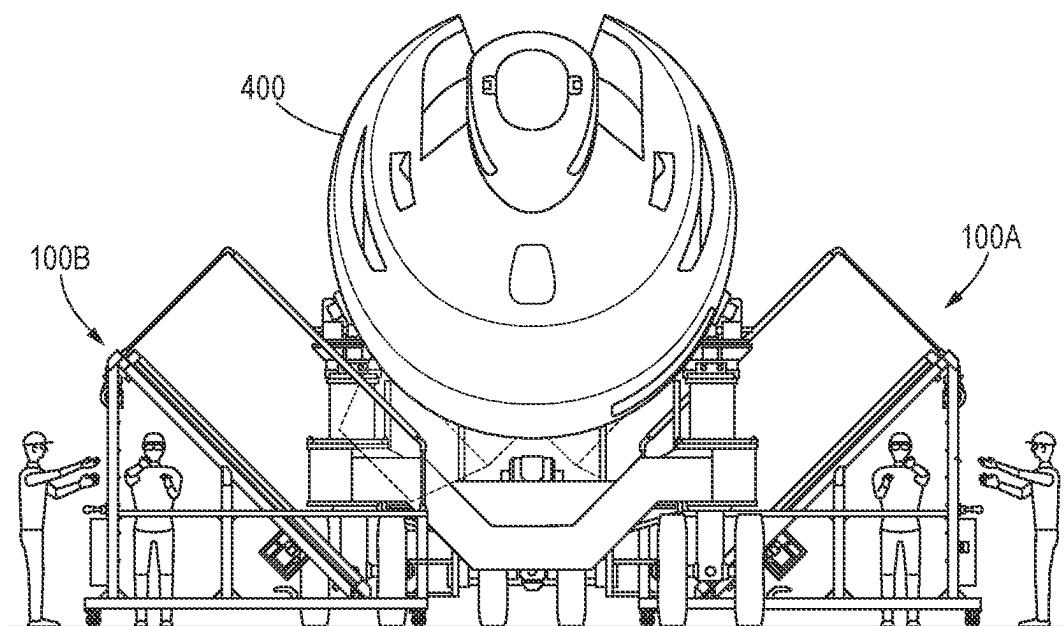


FIG. 13

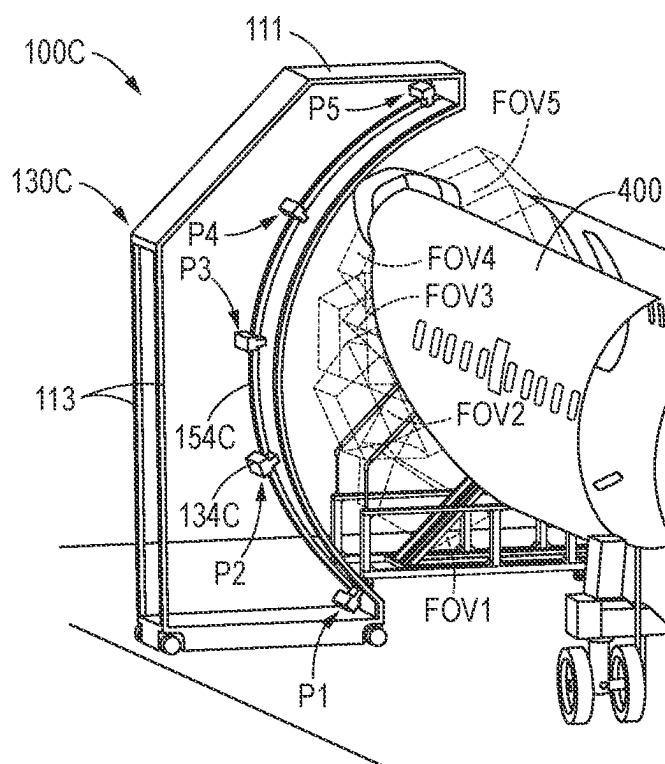


FIG. 14

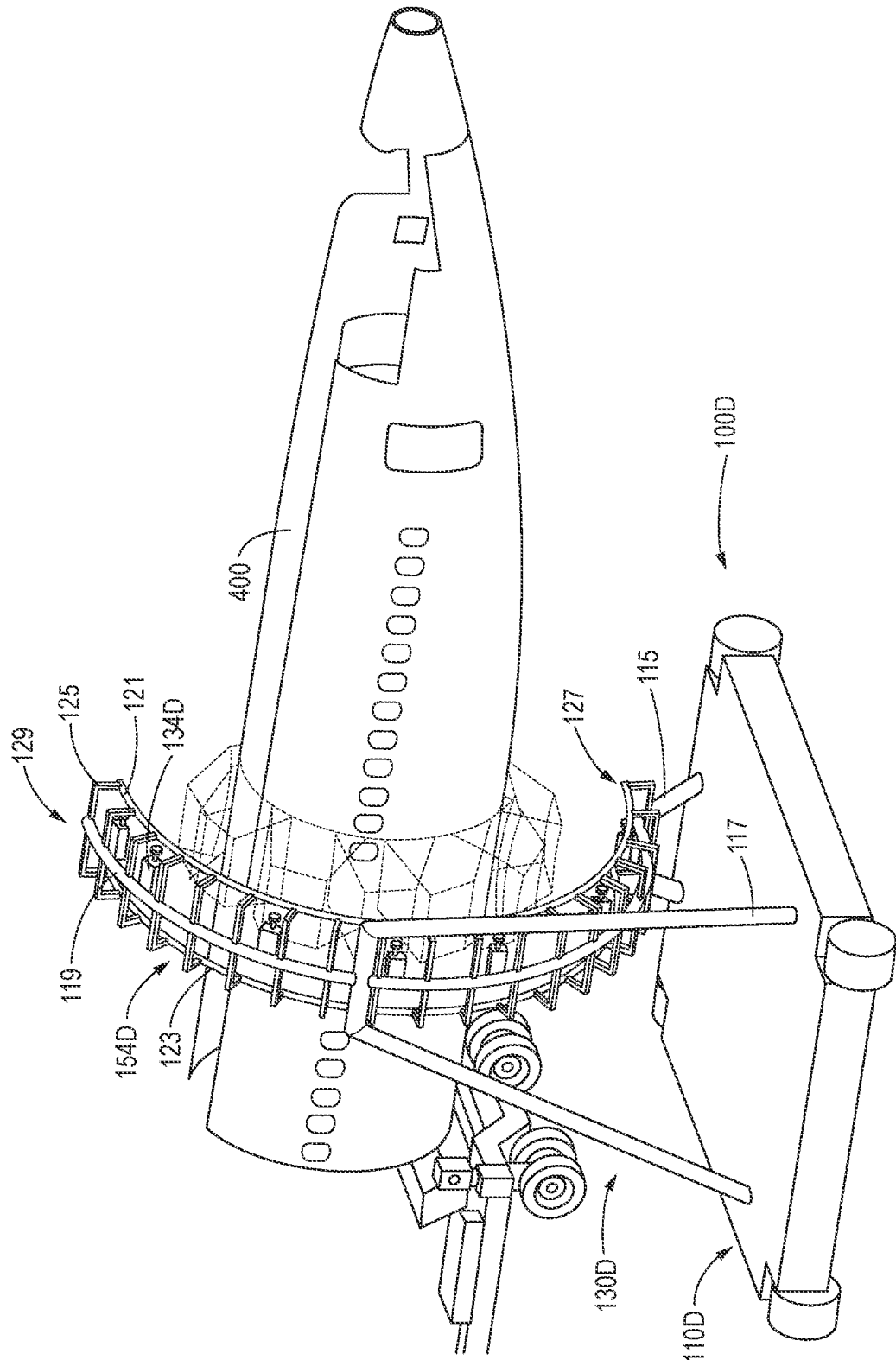


FIG. 15

APPARATUSES AND METHODS FOR LARGE AREA WIRELESS FUSELAGE DENT INSPECTION

FIELD

[0001] Aspects of the present disclosure relate to apparatuses, systems, and methods of inspecting large area objects, such as airplane fuselages.

BACKGROUND

[0002] Conventionally, inspection of airplane fuselages for defects has been a manual process involving flashlights and bump caps, dependent on human eyesight and meticulousness. Accordingly, inspecting an airplane fuselage for defects has traditionally been quite time consuming. Moreover, airplane fuselages are usually inspected during final assembly while the airplane fuselage is coated in a green Temporary Protective Coating (TPC), which can obstruct the view of many defects. Consequently, some defects can go undiscovered until a later stage, such as after a shiny final paint job is applied on an airplane's fuselage. Rework to fix such undetected defects at such a stage can be costly. In some instances, for example, a cockpit or lavatory may need to be removed from the fuselage in order to pull out dents that went undetected during final assembly. Currently-available scanners and vision systems have been of little help in automating the inspection process as they are relatively small, localized, and take too long to process images.

SUMMARY

[0003] The present disclosure provides a method in one aspect. The method includes translating a 3D scanner relative to a target object, the 3D scanner is arranged in an inspection cart; scanning, as the 3D scanner is translated, the target object using the 3D scanner to capture images of at least a slice of the target object, wherein optical fiducials integrated into the inspection cart are arranged in a field of view of the 3D scanner during the scanning; stitching together the captured images of the target object obtained during the scanning to create a slice image, the captured images being stitched together according to position targets provided by the optical fiducials; and detecting one or more defects on the target object based on the slice image.

[0004] In one aspect, in combination with any example method above or below, the method includes projecting, by the 3D scanner, one or more defect indicators onto the target object to indicate respective ones of the one or more defects detected on the target object.

[0005] In one aspect, in combination with any example method above or below, at least one defect indicator of the one or more defect indicators is projected onto the target object outlining a defect of the one or more defects detected on the target object.

[0006] In one aspect, in combination with any example method above or below, the 3D scanner is translated along a linear path that is oriented at an angle relative to a vertical direction.

[0007] In one aspect, in combination with any example method above or below, the optical fiducials include a first fiducial rail and a second fiducial rail that are spaced from one another and arranged parallel to the linear path.

[0008] In one aspect, in combination with any example method above or below, the position targets are arranged on

first fingers that extend from the first fiducial rail and on second fingers that extend from the second fiducial rail, the first fingers are spaced from one another along the first fiducial rail and the second fingers are spaced from one another along the second fiducial rail.

[0009] In one aspect, in combination with any example method above or below, the method includes masking the optical fiducials in the slice image so that the optical fiducials do not appear on the slice image.

[0010] In one aspect, in combination with any example method above or below, the slice of the target object is a first slice, and wherein the method further includes: a) moving the inspection cart so that the inspection cart aligns with a subsequent slice of the target object; b) translating, with the inspection cart aligned with the subsequent slice, the 3D scanner relative to the target object; c) scanning, as the 3D scanner is translated with the inspection cart aligned with the subsequent slice, the target object using the 3D scanner to capture images of the subsequent slice of the target object, wherein the optical fiducials are arranged in the field of view of the 3D scanner during the scanning of the subsequent slice; d) stitching together captured images of the subsequent slice of the target object to create a subsequent slice image, the captured images of the subsequent slice being stitched together according to the position targets provided by the optical fiducials; and e) detecting one or more defects on the subsequent slice of the target object based on the subsequent slice image.

[0011] In one aspect, in combination with any example method above or below, the method further includes iterating a) through e) for a predetermined length of the target object; and stitching the slice image of the first slice and subsequent slice image of each one of the subsequent slices into a combined slice image.

[0012] In one aspect, in combination with any example method above or below, the combined slice image is a first combined slice image associated with a first side of the target object, and wherein the method further includes creating a second combined slice image associated with a second side of the target object.

[0013] In one aspect, in combination with any example method above or below, the method includes creating a digital twin of the target object based at least in part on the first combined slice image and the second combined slice image.

[0014] In one aspect, in combination with any example method above or below, the inspection cart moves autonomously from one slice of the target object to another.

[0015] In one aspect, in combination with any example method above or below, the target object is a fuselage, and wherein the fuselage is scanned so that the captured images of the slice extend from a midline of an underbelly of the fuselage to a window line of the fuselage.

[0016] In one aspect, in combination with any example method above or below, the target object is a fuselage, and wherein the fuselage is scanned so that the captured images of the slice extend from a midline of an underbelly of the fuselage to a midline of a top of the fuselage.

[0017] The present disclosure provides an inspection system in another aspect. The inspection system includes an inspection cart. The inspection cart includes a tower supporting a track. The inspection cart further includes a carriage movable along the track, the carriage supporting a 3D scanner having at least two cameras and a projector.

Further, the inspection cart includes optical fiducials arranged relative to the track so that the optical fiducials are in a field of view of the 3D scanner during scanning of a target object.

[0018] In one aspect, in combination with any example inspection system above or below, the inspection system further includes one or more processors and one or more non-transitory memory devices storing a program, which, when executed by any combination of the one or more processors, causes the one or more processors to perform an operation. The operation includes causing the carriage to move along the track so that the 3D scanner is moved relative to the target object, causing, as the 3D scanner is translated, the 3D scanner to scan the target object to capture images of at least a slice of the target object; stitching together the captured images of the slice of the target object to create a slice image, the captured images being stitched together according to position targets provided by the optical fiducials; and detecting one or more defects on the target object based on the slice image.

[0019] In one aspect, in combination with any example inspection system above or below, the operation further includes causing the projector of the 3D scanner to project one or more defect indicators onto the target object to indicate respective ones of the one or more defects detected on the target object.

[0020] In one aspect, in combination with any example inspection system above or below, the track is a linear track and the optical fiducials include a first fiducial rail and a second fiducial rail that are spaced from one another and arranged parallel to the linear track.

[0021] In one aspect, in combination with any example inspection system above or below, the target object is a fuselage, and wherein the track is a curved track that extends above and below the fuselage.

[0022] The present disclosure provides an inspection cart in yet another aspect. The inspection cart includes a tower supporting a track; a carriage movable along the track, the carriage supporting a 3D scanner having at least two cameras and a projector; and optical fiducials arranged relative to the track so that the optical fiducials are in a field of view of the 3D scanner during scanning of a target object.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] So that the manner in which the above recited features can be understood in detail, a more particular description, briefly summarized above, may be had by reference to example aspects, some of which are illustrated in the appended drawings.

[0024] FIG. 1 is a forward-looking-aft perspective view of an inspection cart according to example aspects of the present disclosure.

[0025] FIG. 2 is an aft-looking-forward perspective view of the inspection cart of FIG. 1.

[0026] FIG. 3 is a side view of the inspection cart of FIG. 1.

[0027] FIG. 4 is a system diagram of an inspection system according to example aspects of the present disclosure.

[0028] FIG. 5 is a flow diagram for an example method of inspecting a target object using an inspection system according to example aspects of the present disclosure.

[0029] FIG. 6 is a schematic side view of a fuselage of an airplane showing one example manner in which the fuselage can be divided into slices.

[0030] FIGS. 7 and 8 depict an inspection cart positioned relative to a slice of a port side of the fuselage and performing a scan thereof.

[0031] FIG. 9 depicts a plurality of images of the slice of the fuselage captured during the scan depicted in FIGS. 7 and 8.

[0032] FIG. 10 shows a combined slice image stitched together from the images of the slice depicted in FIG. 9.

[0033] FIG. 11 shows a defect indicator being projected by a projector of a 3D scanner onto a fuselage.

[0034] FIG. 12 shows a plurality of slice images each associated with one of the predefined slices of a fuselage, wherein the slice images are stitched together into a combined slice image.

[0035] FIG. 13 depicts a forward-looking-aft view of a fuselage and shows a first inspection cart and a second inspection cart scanning and capturing images of respective sides of the fuselage according to example aspects of the present disclosure.

[0036] FIG. 14 is a perspective view of an inspection cart having a siege tower in a scanner/motivator configuration according to example aspects of the present disclosure.

[0037] FIG. 15 is a perspective view of another inspection cart having a siege tower in a rollercoaster-style conveyer configuration according to example aspects of the present disclosure.

DETAILED DESCRIPTION

[0038] The present disclosure relates to improved apparatuses, systems, and methods of inspection of airplane fuselages. Particularly, the apparatuses, systems, and methods disclosed herein facilitate detection of defects on airplane fuselages, e.g., in real time. The apparatuses and methods of the present disclosure can be utilized for inspection of an airplane fuselage at various life cycle stages, such as at final assembly, at delivery, or even after an airplane has been fielded (e.g., for an inspection after a hail storm). Although the inspection apparatuses, systems, and methods are described herein in the context of inspecting airplane fuselages, the aspects of the present disclosure are also applicable to inspecting other large area objects, such as other areas of the airplane (e.g., wings, cockpit, tail empennage, etc.), ships, spacecraft, trains, industrial pipes, etc.

[0039] In one example aspect, an inspection cart of an inspection system is provided that facilitates inspection of airplane fuselages by detecting skin defects, e.g., in real time. The inspection cart is mobile and includes a tower supporting a track. The inspection cart also includes a carriage movable along the track. The carriage supports a 3D scanner having at least two cameras and a projector. The inspection cart also includes integrated optical fiducials arranged relative to the track so that the optical fiducials are in a field of view of the 3D scanner during scanning of the fuselage. In this regard, the optical fiducials are integrated into the inspection cart. During inspection of a fuselage, the 3D scanner can be translated along the track. As the 3D scanner is translated, the 3D scanner captures images of at least a slice of the fuselage. The integrated optical fiducials are arranged in a field of view of the 3D scanner during the scanning. The captured images can be stitched together to create a slice image. The captured images are stitched together according to position targets provided by the optical fiducials. One or more defects on the fuselage can be detected based on the slice image. Detected defects can be

presented to an operator during a scan of a slice or right after a slice has been scanned. Particularly, one or more defect indicators can be projected onto the fuselage to indicate the locations of the defects. This may allow an operator to immediately appreciate the defects and their respective locations on the fuselage. Other alerts or reports identifying the defects are also possible, such as real time presentation of the identified defects on a display, audio alerts, automatically generated reports, etc. This process can be repeated for other slices of the fuselage and the slice images can be stitched together to form a combined slice image. In some aspects, multiple inspection carts can be utilized to scan different sections of a fuselage simultaneously.

[0040] The inspection carts, inspection systems, and methods of the present disclosure can provide certain advantages, benefits, and/or technical effects. For instance, manual inspection of a fuselage to detect defects (e.g., dents, missing parts, unacceptable tolerances, etc.) can be eliminated or largely replaced by the apparatuses and methods provided herein. Eliminating or reducing the need for manual inspection can allow for fewer escapes, lower costs, greater repeatability, precision and accuracy of detected defects, improved quality and safety, reduced inspection time, elimination or reduction of ergonomic challenges to operators (e.g., head injuries on the underside of a fuselage, repetitive stress (awkward postures), eye strain, and noise. Further, the optical fiducials are integrated into the inspection cart, which allows for contactless inspection of a fuselage. Accordingly, the fuselage need not be touched or contacted to arrange optical targets thereon. Optical fiducials have traditionally been placed on a target part or a fixture holding the target part, e.g., with stickers in known positions. The setup and cleanup of this traditional technique would be time consuming and cost prohibitive for large target parts, such as airplane fuselages.

[0041] In addition, defects can be detected and their positions can be made immediately known to an operator, e.g., by projecting defect indicators onto the airplane's fuselage. An operator can use the indicators to mark the aircraft, as desired, e.g., with tape or the like. Moreover, the carriage that carries the 3D scanner can be driven in a controlled motion with the optical fiducials being at a known distance and in the field of view of the cameras. The 3D scanner can use position targets on the "fingers" and/or fiducial rails to determine its relative position, which assists in stitching together captured images for a slice image of the fuselage. Further, the inspection cart is mobile and thus can be moved (e.g., manually or autonomously) from one capture position to the next to capture different slices of the fuselage. The slice images can be stitched together to create a combined slice image, e.g., of the length of the airplane's fuselage. The slice images and/or combined slice image can be used to generate a digital twin of the fuselage, which can be used, among other possible uses, as a baseline for comparison purposes with future scans of the fuselage. In addition, The slice images and/or combined slice image can be used by predictive assembly systems to make assembly predictions, such as predictions as to how to address one or more of the detected defects, e.g., by adding a shim, machining a surface, etc.

[0042] Turning now to the drawings, FIGS. 1, 2, 3 provide various views of an inspection cart 100 according to example aspects of the present disclosure. For reference, the inspection cart 100 defines a transverse direction T, a lateral

direction L, and a vertical direction V, which are mutually perpendicular to one another and form an orthogonal direction system. The inspection cart 100 has a forward end 102 and an aft end 104. The forward end 102 and the aft end 104 are spaced from one another along the transverse direction T. The inspection cart 100 also has a first side 106 and a second side 108. The first side 106 is spaced from the second side 108 along the lateral direction L.

[0043] The inspection cart 100 has a chassis 110 to which a plurality of casters 112 are mounted. The casters 112 enable the inspection cart 100 to be moved. The chassis 110 is generally formed by a plurality of frame members and has a rectangular configuration. The inspection cart 100 also has a first side mount 114 and a second side mount 116, which both are connected to and supported by the chassis 110. The first side mount 114 includes a first side rail 118, a first forward post 120, and a first intermediate post 122. Likewise, the second side mount 116 includes a second side rail 124, a second forward post 126, and a second intermediate post 128. The first and second side rails 118, 124 are each arranged at an ergonomically-friendly height where an adult of average height can easily grasp, and if desired, manually move the inspection cart 100 by pushing or pulling one or both of the first and second side rails 118, 124.

[0044] The inspection cart 100 also has a tower 130 that structurally supports a translating carriage 132 that carries a 3D scanner 134. The tower 130 includes, among other things, a first tower post 136, a second tower post 138, a tower cross member 140 extending between and connecting the first and second tower posts 136, 138 at their top ends. The tower 130 also includes a first angled frame 141 and a second angled frame 142. The first angled frame 141 and the second angled frame 142 are coupled to the tower cross member 140 at a top of the tower 130 and to a chassis cross member 144 at a bottom of the tower 130. The tower 130 also includes a first intermediate tower post 146 and a second intermediate tower post 148 that support the mid-spans of the first and second angled frames 141, 142. The first side rail 118 and the second side rail 124 connect respectively to the first tower post 136 and the second tower post 138. An aft handle bar 150 is connected to the first and second tower posts 136, 138, and offers an ergonomically-friendly bar for a user to grasp and move the inspection cart 100 manually. The aft handle bar 150 also advantageously extends out over a controls cabinet 152, e.g., for providing protection thereto. The controls cabinet 152 can store various power, communication, and computing equipment associated with operating the inspection cart 100.

[0045] The tower 130 also includes a track 154 and a drive system 156 that functions to translate or move the carriage 132 along the track 154. The track 154 includes a first-side rail 158 and a second-side rail 160 coupled respectively with the first and second angled frames 141, 142. The track 154 is a linear track in this example. In this regard, the first-side and the second-side rails 158, 160 are linear. In some example aspects, the first-side and the second-side rails 158, 160 can include end stops at their respective top and/or bottom ends, e.g., to provide a failsafe for the carriage and/or 3D scanner 134 carried therein. As one example, the end stops can be shock absorbers placed at the bottom of the first-side rail 158 and/or the second-side rail 160. As another example, the end stops can be compression springs placed at the bottom of the first-side rail 158 and/or the second-side rail 160. As yet another example, the end stops can be

elastomer stop buffers placed at the top of the first-side rail **158** and/or the second-side rail **160**. An elastomer stop buffer **162** is shown in FIG. 1 being arranged at a top of the second-side rail **160**.

[0046] The drive system **156** is a belt drive system in this example and includes, among other things, an electric motor **164**, a gearbox **166** mechanically coupled with the electric motor **164**, and a belt pulley system **168** mechanically coupled with the gearbox **166**. Generally, the carriage **132** is slidably received within the track **154** and the drive system **156** can be controlled to translate the carriage **132** along the track **154**, and consequently, the 3D scanner **134** as well. The carriage **132** is translated along the track **154** between a first position (e.g., a bottom position) and a second position (e.g., a top position). The carriage **132** is shown in the bottom or first position in FIGS. 1 and 3 while the carriage **132** is shown in the top or second position in FIG. 2. Accordingly, the 3D scanner **134** can be translated along a translating path TP (FIG. 3), which is a linear path that is oriented at an angle relative to the vertical direction V in this example. In some example aspects, the translating path TP can be oriented at a forty-five degree angle (45°) with respect to the vertical direction V. However, the translating path TP can be oriented at other angles, such as thirty degrees (30°) or sixty degrees (60°) in other example aspects.

[0047] The inspection cart **100** also includes optical fiducials that are arranged in a Field Of View (FOV) of the 3D scanner **134** during scanning of a target object, such as an airplane fuselage. In this example, the optical fiducials include a first optical fiducial **170** and a second optical fiducial **172**. The first optical fiducial **170** includes a first fiducial rail **174** and the second optical fiducial **172** includes a second fiducial rail **176**. The first and second fiducial rails **174**, **176** are spaced from one another, e.g., along the lateral direction L, and are arranged parallel to the translating path TP, or rather, parallel to a translating angle of the 3D scanner **134**. The first fiducial rail **174** and the second fiducial rail **176** are both spaced from the track **154**, e.g., along a direction that is perpendicular to the translating path TP. A bottom end of the first fiducial rail **174** is supported by a first forward-support post **178**, which is connected to the first side mount **114** and extends lengthwise along the vertical direction V. A top end of the first fiducial rail **174** is connected to a first aft-support post **180**, which is connected to the tower cross member **140** and extends lengthwise along a direction that is perpendicular to the translating path TP. Similarly, a bottom end of the second fiducial rail **176** is supported by a second forward-support post **182**, which is connected to the second side mount **116** and extends lengthwise along the vertical direction V. A top end of the second fiducial rail **176** is connected to a second aft-support post **184**, which is connected to the tower cross member **140** and extends lengthwise along a direction that is perpendicular to the translating path TP.

[0048] As shown in FIG. 2, the first optical fiducial **170** and the second optical fiducial **172** can include position targets **186**. Generally, the position targets **186** function to assist the 3D scanner **134** in determining its position as it translates during a scan of a target object and also enables captured images to be stitched together to form a combined image. In some aspects, the position targets **186** can be located on the first and second fiducial rails **174**, **176**. Additionally or alternatively, the position targets **186** can be located on fingers of the first and second fiducial rails **174**,

176. In some aspects, for example, the position targets **186** can be arranged on first fingers **188** that extend from the first fiducial rail **174** and on second fingers **190** that extend from the second fiducial rail **176**. The first fingers **188** are spaced from one another along the first fiducial rail **174** and the second fingers **190** are spaced from one another along the second fiducial rail **176**. The first and second fingers **188**, **190** each extend from their respective first and second fiducial rails **174**, **176** toward one another along the lateral direction L. In the depicted example of FIG. 2, the position targets **186** are disposed on each of the first fingers **188** and along an entirety of the length of the first fiducial rail **174**. Moreover, the position targets **186** are disposed on each of the second fingers **190** and along an entirety of the length of the second fiducial rail **176**. The respective close-up views in FIG. 2 depict the position targets **186** on the noted features. The position targets **186** are arranged to face toward or otherwise be in the FOV of the 3D scanner **134**.

[0049] The position targets **186** can be arranged in different patterns along the first and second fiducial rails **174**, **176** and the first and second fingers **188**, **190**. The patterns and locations (e.g., coordinates relative to the 3D scanner **134**) of the position targets **186** can be known to the 3D scanner **134** and/or a computing system associated with the inspection cart **100** so that the position targets **186** can be utilized to assist the 3D scanner **134** in determining its position as it translates during a scan of a target object and also for stitching captured images together. Advantageously, the first and second optical fiducials **170**, **172** are integrated into the inspection cart **100** and have position targets **186** arranged in the FOV of the 3D scanner **134**. In this way, the 3D scanner **134** can know or determine its position relative to the position targets **186**—without any targets or markups being needed on the target object itself. This provides non-contact scanning capability.

[0050] FIGS. 1, 2, and 3 provide an inspection cart according to an example aspect of the present disclosure. In some alternative aspects, the inspection cart **100** of FIGS. 1, 2, and 3 can have other configurations. For example, in some aspects, the inspection cart **100** can be arranged on wheels that lock into a track, e.g., wherein the track extends along at least a portion of the target object, such as a full longitudinal length of a fuselage. The inspection cart **100** can be moved along the track to capture images of various “slices” of the fuselage. In yet other aspects, the inspection cart **100** can be arranged to have one or more steered-wheels and, optionally, one or more non-steered wheels. In such aspects, the inspection cart can include a steering device (e.g., a steering wheel, levers for zero-turn capability, etc.) and can have a motor or engine to propel the inspection cart **100**. In this regard, the inspection cart **100** can be driven to desired locations for scanning. In some further aspects, the inspection cart **100** can include self-driving or autonomous driving features that allow the cart to be moved autonomously, e.g., to various positions along a longitudinal length of a fuselage. In yet further aspects, as will be explained in further detail herein, the inspection cart **100** can include a “siege tower”, e.g., that enables a top quadrant of a fuselage to be scanned in addition to a bottom quadrant.

[0051] With reference now to FIG. 4, in some aspects, the inspection cart **100** of FIGS. 1, 2, 3 can be a part of an inspection system **200**. The inspection system **200** can include the inspection cart **100** (FIGS. 1-3) and a computing system **210**, and can also include a data repository **238**

and/or a predictive assembly system **240** in some aspects. Other systems can also be communicatively coupled with the inspection system **200**, e.g., via a communication bus **244**.

[0052] As depicted in FIG. 4, the inspection cart **100** (FIGS. 1-3) includes the 3D scanner **134** that is translatable along the track **154** (FIG. 1). The 3D scanner **134** includes, among other things, at least two cameras and a projector **191**. In this example, the cameras include a first camera **192** and a second camera **193**, which are arranged on opposite sides of the projector **191**. The first and second cameras **192**, **193** capture images of a target object as the 3D scanner **134** is translated along the track **154**. As will be explained in greater detail herein, during a scan, the projector **191** can project light onto the target object to enhance the images captured by the first and second cameras **192**, **193**. Also, the projector **191** can be used to project defect indicators onto the target object to indicate respective detected defects. The 3D scanner **134** also includes a first laser emitter **194** and a second laser emitter **195**, which are spaced from one another. The first and second laser emitters **194**, **195** can emit lasers onto the target object, e.g., to ensure focal distances between the first and second cameras **192**, **193** and the target object are within a predetermined range or specification. For instance, when lasers emitted respectively from the first and second laser emitters **194**, **195** meet at a point, the focal distances can be determined to be in the predetermined range or specification. In this regard, the first and second laser emitters **194**, **195** can be used to determine the inspection cart **100** has a suitable “standoff” with respect to the target object.

[0053] The 3D scanner **134** can be communicatively coupled with other systems onboard the inspection cart **100** (FIGS. 1-3), such as components within the controls cabinet **152** (FIG. 1) and/or onboard components of the computing system **210**, and also to one or more offboard systems, such as to offboard components of the computing system **210**, the data repository **238**, the predictive assembly system **240**, etc. In some aspects, the computing system **210** can be positioned in part onboard and in part offboard of the inspection cart **100**. In other example aspects, the computing system **210** can be positioned entirely onboard the inspection cart **100**.

[0054] The computing system **210** can include one or more processors **212** and one or more non-transitory memory devices **214** that can store computer-readable instructions **216** or code that can be executed by the one or processors **212** to perform an operation, such as causing the first and second cameras **192**, **193** to capture images of a target object as the 3D scanner **134** is moved along the track **154**, stitching together captured images, and/or causing the projector **191** to project defect indicators onto the target object that correspond to detected defects. The instructions **216** can be software written in any suitable programming language or can be implemented in hardware. The one or more processors **212** and one or more non-transitory memory devices **214** can be embodied in one or computing devices, for example.

[0055] The one or more processor(s) **212** can include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, logic device, or other suitable processing device. The one or more memory device(s) **214** can include one or more computer-readable medium, including, but not limited to, non-transitory computer-readable

medium, RAM, ROM, hard drives, flash drives, and other memory devices. The one or more memory device(s) **214** can store information accessible by the one or more processor(s) **212**, including the instructions **216** that can be executed by the one or more processor(s) **212**. The memory device(s) **214** can further store data **218** that can be accessed by the processors **212**. For example, the data **218** can include images captured by the 3D scanner **134**, known coordinates of the position targets **186** (FIG. 2), data relating to the target object, etc. The data **218** can include one or more table(s), function(s), algorithm(s), model(s), equation(s), etc. according to example aspects of the present disclosure.

[0056] The computing system **210** also include a communication interface **220** used to communicate, for example, with the other components of the inspection system **200**. The communication interface **220** can include any suitable components for interfacing with one or more network(s), including for example, transmitters, receivers, ports, controllers, antennas, or other suitable components.

[0057] As further shown in FIG. 4, in some example aspects, the instructions **216** can include an image analysis module **222** that includes an image stitching container **224** and a defect detection container **226**. The one or more processor(s) **212** of the computing system **210** can receive a plurality of captured images **228** from the 3D scanner **134** and can stitch the captured images together to create a slice image that represents a “slice” of the large area target object. Slice images can be merged or stitched together with other slice images of the target object to ultimately render a combined slice image of the target object. When the one or more processor(s) **212** of the computing system **210** execute the defect detection container **226**, one or more defects on the target object can be detected in the slice image and/or combined slice image and/or merged image. In at least some examples, the coordinates of the defects can be determined and defect indicators **230** can be generated corresponding to the coordinates of the detected defects. The computing system **210** can cause the projector **191** of the 3D scanner **134** to project the defect indicators **230** onto the target object, e.g., with cross hairs, arrows, circles, etc. In some aspects, at least one defect indicator can be projected onto the target object outlining a defect. That is, the projected defect indicator can outline a perimeter of the defect, which can make an area of a defect readily apparent to an operator reviewing the defects on the target object.

[0058] Furthermore, in some aspects, the instructions **216** can include a digital twin creator **232**. When the one or more processors of the computing system **210** execute the digital twin creator **232**, a digital twin **234** of the target object can be created, e.g., with the defects noted on the target object. The digital twin **234** can digitally represent a portion of the target object (e.g., a lower quadrant of an airplane fuselage) or an entirety of a target object (e.g., an entirety of an airplane fuselage). The digital twin **234** can be used for a number of purposes, such as providing a baseline assessment of the target object at final assembly, which can be compared to future scans of the target object (e.g., after rework, at delivery, after being fielded, etc.). Moreover a digital twin **234** can be used for a number of other reasons.

[0059] In some additional aspects, the instructions **216** can include an autonomous movement module **236**. In such aspects, the inspection cart **100** (FIGS. 1-3) can be an autonomous vehicle. As one example, the inspection cart **100** can be a self-driving cart that can be driven about

autonomously, or without human intervention. As another example, the inspection cart **100** can be a self-guided cart that can move itself autonomously along a track, e.g., from one position to another. The autonomous movement module **236** can include logic needed to move the inspection cart **100** from one scan position to another, e.g., based on sensor feedback arranged on the inspection cart **100**, such as sensor feedback from cameras positioned onboard the inspection cart **100**.

[0060] The data repository **238** can store various captured images and/or stitched images of the target object, the digital twin **234**, etc. Such features can be stored in memory for future use and analysis. The predictive assembly system **240** can be communicatively coupled with the other elements of the inspection system **200**. The predictive assembly system **240** can call or receive captured and/or stitched images, the digital twin **234**, etc., and with an appreciation of the detected defects, the predictive assembly system **240** can provide assembly predictions **242**, such as a 3D geometry of a shim that can be placed at a joint between two interfacing components to provide enhanced mechanical properties to the joint. In this regard, an integrated approach to inspection and assembly can be achieved.

[0061] With the inspection system **200** of FIG. **4** generally described, an example manner in which the inspection system **200** can be used to inspect an airplane fuselage for defects will be provided below.

[0062] FIG. **5** is a flow diagram for an example method **300** of inspecting a target object using an inspection system. The method **300** will be explained in the context of the inspection system **200** of FIG. **4** inspecting an airplane fuselage. Accordingly, FIGS. **1** through **4** may also be referenced below in addition to FIG. **5**. However, as will be appreciated, the method **300** can be implemented by inspection systems and inspection carts having different configurations than the inspection system **200** and the inspection cart **100** of FIGS. **1** through **4**.

[0063] At **302**, the method **300** can include commencing a scan of at least a slice of a target object, which is an airplane fuselage in this example. Some airplane fuselages can be quite large in area relative to the FOV of the 3D scanner **134**, and accordingly, the longitudinal length of an airplane fuselage can be scanned slice-by-slice. That is, an airplane fuselage can be divided into a plurality of slices along its longitudinal length. For example, FIG. **6** is a schematic side view of a fuselage **400** of an airplane showing one example manner in which the fuselage **400** can be divided into slices. The fuselage **400** has a forward end **402** and an aft end **404** and extends along a longitudinal direction **L1**. As illustrated, the fuselage **400** is divided into a plurality of slices **406**, or in this example, slices **S1-S14**, which represents fourteen (14) slices. The slices **406** are divided along the longitudinal direction **L1**. The slices **406** can cover the full longitudinal length of the fuselage **400** or a portion thereof. However, in other example aspects, the slices **406** can be divided along a portion of the longitudinal length of the fuselage **400**.

[0064] Generally, the inspection cart **100** can be positioned relative to a given slice of the fuselage **400**, the given slice can be scanned by the 3D scanner **134** to capture images of the given slice of the fuselage **400**, and then the inspection cart **100** can be moved or repositioned relative to a subsequent slice (e.g., an adjacent slice) of the fuselage **400**. With the inspection cart **100** positioned relative to the subsequent slice, the subsequent slice can be scanned by the

3D scanner **134** to capture images of the subsequent slice, and then repositioned relative to another subsequent slice. This process may iterate, e.g., until a full longitudinal length of the fuselage **400** has been scanned, e.g., slice-by-slice. In at least some example aspects, the slices **406** are defined with a smaller width than the FOV of the 3D scanner **134**, e.g., so that the captured images of the slices (or slice images) overlap, which can be advantageous for stitching the slice images together into a combined slice image. As at least a part of the method **300** can be iterative (e.g., the scanning aspect), a scan of a subsequent slice of the target object can also be commenced at **302**.

[0065] At **304**, the method **300** can include arranging the inspection cart of the inspection system relative to the target object. For instance, the inspection cart **100** can be arranged relative to a given slice of the fuselage **400** to be scanned. The inspection cart can be moved manually or the inspection cart **100** can move itself autonomously. In some example aspects, the inspection cart **100** can be arranged relative to the fuselage **400** by performing a “standoff process” to ensure the inspection cart **100** is at a suitable distance from the fuselage **400**. In an example standoff process, lasers can be emitted by the first and second laser emitters **194**, **195** of the 3D scanner **134**, and when the lasers meet at a point or within a predetermined distance of one another, the focal distances between the first and second cameras **192**, **193** and the fuselage **400** can be determined to be in a predetermined range or specification. This can ensure that the cameras are positioned in a satisfactory position to capture images of the fuselage **400** during a scan. In at least some aspects, the standoff process is performed for each slice of the fuselage **400** with the 3D scanner located in a same position along the track **154**, e.g., a top position or a bottom position. This can facilitate continuity of the captured images and can facilitate stitching together of slice images into a combined slice image. Once the inspection cart **100** is arranged relative to a slice of the fuselage **400**, e.g., at a suitable standoff, scanning of the slice can commence.

[0066] At **306**, the method **300** can include translating a 3D scanner relative to a target object. For instance, once the inspection cart **100** is arranged relative to a slice of the fuselage **400** at **304**, the 3D scanner **134** can be translated, e.g., along the translation path **TP**. Specifically, the electric motor **164** can be activated to drive the gearbox **166**, which in turn drives a belt of the belt pulley system **168**. When the belt is driven, the carriage **132** that carries the 3D scanner **134** can be moved along the track **154**, or rather, along the translation path **TP**. Accordingly, the 3D scanner **134** is translated. The 3D scanner **134** can start at one end of the track **154** and translate to the other end of the track, for example. Alternatively, the 3D scanner **134** can be moved along the translation path **TP** from a predetermined starting position to a predetermined ending position.

[0067] At **308**, the method **300** can include scanning, as the 3D scanner is translated at **306**, a target object using the 3D scanner to capture images of at least a slice of the target object, wherein optical fiducials integrated into the inspection cart are arranged in a FOV of the 3D scanner during the scanning. For instance, the projector **191** of the 3D scanner **134** can project light onto the fuselage **400**. With the light projected onto fuselage **400**, the first and second cameras **192**, **193** can capture images of at least a slice of the fuselage **400** as the 3D scanner **134** is translated. A plurality of images can be captured for a given slice. Further, as the first and

second cameras **192**, **193** capture the images, the first and second fiducial rails **174**, **176** and the first and second fingers **188**, **190** are arranged in the FOV of the 3D scanner **134**, and as a result, these items are present in the captured images.

[0068] By way of example, FIG. 7 depicts the inspection cart **100** positioned relative to a slice of a port side of the fuselage **400** (e.g., the third slice S3 of the fuselage **400** in FIG. 6). As shown in FIG. 7, the projector **191** of the 3D scanner **134** is projecting light onto the underbelly of the fuselage **400** and the 3D scanner **134** is positioned at a bottom position along the track **154** (FIG. 1). The FOV of the first and second cameras **192**, **193** is labeled in FIG. 7 as “FOV”. Notably, the optical fiducials **170**, **172** are arranged within the FOV, and therefore, the position targets **186** (FIG. 2) are also in the FOV. The first and second cameras **192**, **193** can capture images of the fuselage **400** as the 3D scanner **134** is translated along the track **154**, or rather, the translation path TP (FIG. 3). The 3D scanner **134** can capture images of the fuselage **400**, e.g., at a predefined interval, upon sensing a given position target, etc. Eventually the 3D scanner **134** reaches an end of the track **154** or a predefined position. As shown in FIG. 8, the 3D scanner **134** is shown at a top position along the track **154**. In this regard, the 3D scanner **134** has traversed from the bottom position (FIG. 7) to the top position (FIG. 8) along the translation path TP capturing images along the way. In FIG. 8, the FOV is at or just above a window line WL of the fuselage **400**, which extends along the longitudinal direction L1 defined by the fuselage **400**. Accordingly, for the scan of the slice of the fuselage **400** in FIGS. 7 and 8, the fuselage **400** is scanned so that the captured images of the slice extend from a midline of an underbelly of the fuselage **400** to the window line WL of the fuselage **400**. In this regard, the scanned slice is a slice of the port side, bottom quadrant of the fuselage **400**.

[0069] As the 3D scanner **134** is translated along the track **154** as shown in FIGS. 7 and 8, images of the fuselage **400** are captured by the first and second cameras **192**, **193**. The captured images **228** are depicted in FIG. 9. Particularly, with the 3D scanner **134** at a first position, e.g., at the bottom of the track **154** as shown in FIG. 7, a first image C1 of the fuselage **400** can be captured. Then, with the 3D scanner **134** at a second position along the track **154**, which is different than the first position, a second image C2 of the fuselage **400** can be captured. Next, with the 3D scanner **134** at a third position along the track **154**, which is different than the first and third positions, a third image C3 of the fuselage **400** can be captured. This process can continue for subsequent image captures, e.g., for a fourth image C4, a fifth image C5, a sixth image C6, a seventh image C7, an eighth image C8, and a ninth image C9. The ninth image C9 can be captured by the first and second cameras **192**, **193** when the 3D scanner **134** is positioned at the top of the track **154**, e.g., as shown in FIG. 8. In the captured images C1-C9, the first and second fiducial rails **174**, **176** and first and second fingers **188**, **190** thereof are depicted, and as a result, the position targets **186** are also in the captured images C1-C9. The position targets **186** can be used for stitching the captured images **228** together at **310**. In other aspects, more or less than nine (9) images can be captured during a scan of a slice of the fuselage **400**.

[0070] At **310**, the method **300** can include stitching together the captured images of the target object obtained during the scanning to create a slice image, the captured

images being stitched together according to position targets provided by the optical fiducials. For instance, the one or more processors **212** of the computing system **210** can execute the image stitching container **224** so that the captured images **228**, e.g., images C1-C9 in FIG. 9, are stitched together. The arrangement of the position targets **186** within the captured images **228** can be identified so that the position of the 3D scanner **134** can be determined. As noted previously, the position targets **186** can be uniquely arranged on the first and second fiducial rails **174**, **176** and/or the first and second fingers **188**, **190** (e.g., in unique patterns) so that their locations can be easily identified, which can facilitate quick processing of the position of the 3D scanner **134**. Based on the known position of the 3D scanner **134**, the captured images **228** can be stitched together to form a slice image SL3 as shown in FIG. 10. The slice image SL3 can represent the full image of the third slice S3, for example. Notably, the first and second optical fiducials **170**, **172** and first and second fingers **188**, **190** can be removed from or masked in the slice image SL3. For instance, as stereo vision is obtained through multiple positions, the first and second optical fiducials **170**, **172** and first and second fingers **188**, **190** can be filtered out of the slice image SL3.

[0071] At **312**, the method **300** can include detecting one or more defects on the target object based on the slice image. For instance, the one or more processors **212** of the computing system **210** can execute the defect detection container **226** so that the slice of the fuselage **400** in the slice image SL3 of FIG. 10 can be analyzed for defects. As shown in FIG. 10, for this example, defects **246** have been detected for this slice of the fuselage **400**, including a first defect **246A** and a second defect **246B**. In executing the defect detection container **226**, the one or more processors **212** can ignore known features so as not to identify them as defects. For instance, in executing the defect detection container **226**, the one or more processors **212** can receive data indicating positions of known features in the fuselage **400** that may appear as defects, such as window cutouts, door cutouts, rivets, etc. The data can be processed to ensure that such known features are not identified as defects.

[0072] At **314**, the method **300** can include projecting, by the 3D scanner, one or more defect indicators onto the target object to indicate respective ones of the one or more defects detected on the target object. In at least some examples, the coordinates of the defects detected **246** can be determined and defect indicators **230** can be generated corresponding to the coordinates of the detected defects **246**. The computing system **210** can cause the projector **191** of the 3D scanner **134** to project the defect indicators **230** onto the target object, e.g., with cross hairs, arrows, circles, etc. In some aspects, the defects **246** can be classified into one of a plurality of defect types, e.g., defect type 1, defect type 2, etc., and the defect indicators projected onto the fuselage **400** can be projected with a color associated with the defect type to which the defect has been classified. In some aspects, at least one defect indicator can be projected onto the fuselage **400** outlining a defect. Stated another way, the projected defect indicator can outline a perimeter of the defect, which can make an area of a defect readily apparent to an operator reviewing the defects on the fuselage **400**. By way of example, FIG. 11 shows a first defect indicator **230A** being projected by the projector **191** of the 3D scanner **134** onto the fuselage **400**. The first defect indicator **230A** projected onto the fuselage **400** corresponds to the first

detected defect **246A** illustrated in FIG. **10**. The first defect indicator **230A** outlines a perimeter of the first detected defect **246A**, making an area of the first detected defect **246A** readily apparent to an operator. The 3D scanner **134** can be translated along the track **154** so that defect indicators **230** corresponding to each one of the detected defects **246** can be projected onto the fuselage **400**.

[0073] In some aspects, the operations **312** and **314** can be completed at a later stage, such as after each slice of the fuselage **400** has been scanned and the slice images stitched together into a combined slice image.

[0074] At **316**, the method **300** can include determining whether each relevant slice of the target object has been scanned, e.g., according to operations **302** through **314**. When each relevant slice has not been scanned, the method **300** can iterate to **302** so that a subsequent slice or slices can be scanned. When each relevant slice has been scanned, the method **300** can proceed to **318**. For instance, when slices **S1-S14** (FIG. **6**) of the fuselage **400** have been scanned, or from nose to tail, the method **300** can proceed to **318**. Accordingly, operations **302** through **314** can be iterated for a predetermined length of the target object, e.g., a full longitudinal length of the fuselage **400**.

[0075] At **318**, the method **300** can include stitching the slice images into a combined slice image. For instance, the one or more processors **212** of the computing system **210** can stitch the slice images together into a combined slice image. By way of example, FIG. **12** shows a plurality of slice images **252** each associated with one of the predefined slices of the fuselage **400**. Slice images **SL1**, **SL3**, **SL4**, **SL5**, and **SL14** associated with slices **S1**, **S3**, **S4**, **S5**, and **S14** are shown in FIG. **12** but it will be appreciated that slice images **SL1** through **SL14** can be stitched together to form the combined slice image **254**. In at least some example aspects, the slices are defined with a smaller width than the FOV of the 3D scanner **134**, e.g., so that the slice images overlap, which can be advantageous for stitching the slice images together into a combined slice image.

[0076] At **320**, the method **300** can include creating a second combined slice image associated with the second side of the target object, wherein the combined slice image created at **318** is a first combined slice image associated with a first side of the target object. For instance, the combined slice image **254** can be a first combined slice image **254** associated with a first side of the fuselage **400** (e.g., a port side). A second combined slice image associated with a second side of the fuselage **400** (e.g., a starboard side) can be created, e.g., using operations **302** through **318**. The second combined slice image can be generated having a same or similar shape as the fuselage **400** in the first combined slice image **254** presented in FIG. **12**, except the second combined slice image will identify defects, if any, associated with the second side of the fuselage **400**.

[0077] At **322**, the method **300** can include storing the combined slice image and/or combined slice images in a data repository. For instance, the first combined slice image **254** (and in some instances the second combined slice image and others) can be stored in the data repository, e.g., on one more non-transitory memory devices thereof. The images can be stored for analysis purposes. For example, the predictive assembly system **240** can call or receive captured and/or stitched images, and with defects identified, the predictive assembly system **240** can provide one or more assembly predictions **242**. An assembly prediction can pro-

vide an instruction for how to fix a detected defect, such as by providing instructions for how to machine a surface, build a shim for strengthening a joint, fasten a rivet according to specification, how to rework a particular area to remove a dent, etc. In this regard, an integrated approach to inspection and assembly can be achieved.

[0078] At **324**, the method **300** can include creating a digital twin of the target object based at least in part on the first combined slice image, and in some instances, one or more other combined slice images. For instance, the one or more processors **212** of the computing system **210** can execute the digital twin creator **232** to create a digital twin **234** of the fuselage **400**. The created digital twin **234** can identify the detected defects on the fuselage **400**. The digital twin **234** can be used for a number of purposes, such as providing a baseline assessment of the fuselage **400** at final assembly, which can be compared to future scans of the fuselage **400** (e.g., after rework, at delivery, after being fielded, etc.). The combined slice image or images can be called from the data repository **238** or the memory devices **214**, for example.

[0079] In some implementations, to facilitate additional efficiency in inspecting a fuselage, the inspection system **200** can include at least one inspection cart associated with a first side (e.g., a port side) of the fuselage **400**, and at least one inspection cart associated with a second side (e.g., a starboard side) of the fuselage **400**. For instance, FIG. **13** depicts a forward-looking-aft view of the fuselage **400** and shows a first inspection cart **100A** scanning and capturing images of various slices of a port side of the fuselage **400** to detect defects thereon and also a second inspection cart **100B** scanning and capturing images of various slices of a starboard side of the fuselage **400** to detect defects thereon. The first and second inspection carts **100A**, **100B** can be configured and operate in a similar manner as the inspection cart **100** described herein. In some aspects, the first and second inspection carts **100A**, **100B** can perform aspects of the method **300** simultaneously, which can provide additional efficiency for inspecting the fuselage **400** for defects.

[0080] In yet other implementations, an inspection cart of the inspection system **200** can have a "siege tower" that enables a fuselage to be scanned so that the captured images of a given slice extend at least from a midline of an underbelly of the fuselage to a midline of a top of the fuselage.

[0081] By way of example, FIG. **14** is a perspective view of an inspection cart **100C** having a siege tower in a scanner/motivator configuration. The inspection cart **100C** includes a tower **130C** that supports a track **154C** along which a 3D scanner **134C** can translate. The 3D scanner **134C** can be carried by a carriage, for example. The track **154C** is non-linear, or in this example, curved complementary to the shape of the fuselage **400**. In this regard, the translating path of the 3D scanner **134C** is also curved or non-linear. The track **154C** is connected at its top end by a roof **111**. The roof **111** is supported by aft posts **113**. Although not shown in FIG. **14**, curved optical fiducial rails shaped complementary to the track **154C** can be provided so that position targets are within the FOV of the 3D scanner **134C**.

[0082] In FIG. **14**, a sequence showing one example manner in which the 3D scanner **134C** can be translated is illustrated. In a first position **P1**, the 3D scanner **134C** has a first field of view **FOV1** and captures an image of a portion

of the underbelly of the fuselage 400. In a second position P2, the 3D scanner 134C has a second field of view FOV2 and captures an image of a lower quadrant of the fuselage 400. In a third position P3, the 3D scanner 134C has a third field of view FOV3 and captures an image of a side of the fuselage 400. In a fourth position P4, the 3D scanner 134C has a fourth field of view FOV4 and captures an image of an upper quadrant of the fuselage 400. In a fifth position P5, the 3D scanner 134C has a fifth field of view FOV5 and captures an image of a top side of the fuselage 400. Accordingly, the inspection cart 100C is arranged to capture images of a given slice of the fuselage 400 at least from a midline of an underbelly of the fuselage 400 to a midline of a top of the fuselage 400. In this regard, a slice image created from the images can be representative of a lower quadrant and upper quadrant of a side of the fuselage 400. This scan range can facilitate additional efficiency in inspecting a fuselage and can allow for detection of defects on both the lower and upper quadrants of a fuselage.

[0083] FIG. 15 is a perspective view of an inspection cart 100D having a siege tower in a rollercoaster-style conveyer configuration. The inspection cart 100D includes a tower 130D that supports a track 154D along which a 3D scanner 134D can translate. The 3D scanner 134D is shown in different positions along the track 154D in FIG. 15. The 3D scanner 134D can be carried by a carriage, for example. The track 154D is non-linear, or in this example, curved complementary to the shape of the fuselage 400. In this regard, the translating path of the 3D scanner 134D is also curved or non-linear. The track 154D is supported by a forward brace 115 and by an aft brace 117. The forward brace 115 and the aft brace 117 are supported by a chassis 110D. The track 154D has a backbone 119, first and second load rails 121, 123 along which the carriage or 3D scanner 134D translates, and load rails supports 125 that connect the first and second load rails 121, 123 with the backbone 119 and provide support thereto. The track 154D is uniquely arranged to accommodate relatively large fuselages, and accordingly, when the inspection cart 100D is in position relative to the fuselage 400, a lower end 127 of the track 154D is arranged under the fuselage 400 and the upper end 129 is arranged above the fuselage 400. In this regard, the track 154D “wraps” around a part of the fuselage 400. Although not shown in FIG. 15, curved optical fiducial rails shaped complementary to the track 154D can be provided so that position targets are within the FOV of the 3D scanner 134D.

[0084] The inspection cart 100D is arranged to capture images of a given slice of the fuselage 400 at least from a midline of an underbelly of the fuselage 400 to a midline of a top of the fuselage 400. In this regard, a slice image created from the images can be representative of a lower quadrant and an upper quadrant of a side of the fuselage 400. This scan range can facilitate additional efficiency in inspecting a fuselage and can allow for detection of defects on both the lower and upper quadrants of a fuselage. In at least some aspects, the inspection cart 100D can be an Automated Guided Vehicle (AGV). In such aspects, the inspection cart 100D can move itself autonomously from one scan position to the next.

[0085] In the current disclosure, reference is made to various aspects. However, it should be understood that the present disclosure is not limited to specific described aspects. Instead, any combination of the following features and elements, whether related to different aspects or not, is

contemplated to implement and practice the teachings provided herein. Additionally, when elements of the aspects are described in the form of “at least one of A and B,” it will be understood that aspects including element A exclusively, including element B exclusively, and including element A and B are each contemplated. Furthermore, although some aspects may achieve advantages over other possible solutions and/or over the prior art, whether or not a particular advantage is achieved by a given aspect is not limiting of the present disclosure. Thus, the aspects, features, aspects and advantages disclosed herein are merely illustrative and are not considered elements or limitations of the appended claims except where explicitly recited in a claim(s).

[0086] As will be appreciated by one skilled in the art, aspects described herein may be embodied as a system, method or computer program product. Accordingly, aspects may take the form of an entirely hardware aspect, an entirely software aspect (including firmware, resident software, micro-code, etc.) or an aspect combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.” Furthermore, aspects described herein may take the form of a computer program product embodied in one or more computer readable storage medium(s) having computer readable program code embodied thereon.

[0087] Program code embodied on a computer readable storage medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

[0088] Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The program code may execute entirely on the user’s computer, partly on the user’s computer, as a stand-alone software package, partly on the user’s computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user’s computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

[0089] Aspects of the present disclosure are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatuses (systems), and computer program products according to aspects of the present disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the block(s) of the flowchart illustrations and/or block diagrams.

[0090] These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other device to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the block(s) of the flowchart illustrations and/or block diagrams.

[0091] The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process such that the instructions which execute on the computer, other programmable data processing apparatus, or other device provide processes for implementing the functions/acts specified in the block(s) of the flowchart illustrations and/or block diagrams.

[0092] The flowchart illustrations and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various aspects of the present disclosure. In this regard, each block in the flowchart illustrations or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order or out of order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

[0093] While the foregoing is directed to aspects of the present disclosure, other and further aspects of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

What is claimed is:

1. A method, comprising:

translating a 3D scanner relative to a target object, the 3D scanner is arranged in an inspection cart;

scanning, as the 3D scanner is translated, the target object using the 3D scanner to capture images of at least a slice of the target object, wherein optical fiducials integrated into the inspection cart are arranged in a field of view of the 3D scanner during the scanning;

stitching together the captured images of the target object obtained during the scanning to create a slice image, the captured images being stitched together according to position targets provided by the optical fiducials; and detecting one or more defects on the target object based on the slice image.

2. The method of claim 1, further comprising:

projecting, by the 3D scanner, one or more defect indicators onto the target object to indicate respective ones of the one or more defects detected on the target object.

3. The method of claim 2, wherein at least one defect indicator of the one or more defect indicators is projected onto the target object outlining a defect of the one or more defects detected on the target object.

4. The method of claim 1, wherein the 3D scanner is translated along a linear path that is oriented at an angle relative to a vertical direction.

5. The method of claim 4, wherein the optical fiducials include a first fiducial rail and a second fiducial rail that are spaced from one another and arranged parallel to the linear path.

6. The method of claim 5, wherein the position targets are arranged on first fingers that extend from the first fiducial rail and on second fingers that extend from the second fiducial rail, the first fingers are spaced from one another along the first fiducial rail and the second fingers are spaced from one another along the second fiducial rail.

7. The method of claim 1, further comprising:

masking the optical fiducials in the slice image so that the optical fiducials do not appear on the slice image.

8. The method of claim 1, wherein the slice of the target object is a first slice, and wherein the method further comprises:

a) moving the inspection cart so that the inspection cart aligns with a subsequent slice of the target object;

b) translating, with the inspection cart aligned with the subsequent slice, the 3D scanner relative to the target object;

c) scanning, as the 3D scanner is translated with the inspection cart aligned with the subsequent slice, the target object using the 3D scanner to capture images of the subsequent slice of the target object, wherein the optical fiducials are arranged in the field of view of the 3D scanner during the scanning of the subsequent slice;

d) stitching together captured images of the subsequent slice of the target object to create a subsequent slice image, the captured images of the subsequent slice being stitched together according to the position targets provided by the optical fiducials; and

e) detecting one or more defects on the subsequent slice of the target object based on the subsequent slice image.

9. The method of claim 8, further comprising:

iterating a) through e) for a predetermined length of the target object; and

stitching the slice image of the first slice and subsequent slice image of each one of the subsequent slices into a combined slice image.

10. The method of claim 9, wherein the combined slice image is a first combined slice image associated with a first side of the target object, and wherein the method further comprises:

creating a second combined slice image associated with a second side of the target object.

11. The method of claim 10, further comprising:

creating a digital twin of the target object based at least in part on the first combined slice image and the second combined slice image.

12. The method of claim 8, wherein the inspection cart moves autonomously from one slice of the target object to another.

13. The method of claim 1, wherein the target object is a fuselage, and wherein the fuselage is scanned so that the

captured images of the slice extend from a midline of an underbelly of the fuselage to a window line of the fuselage.

14. The method of claim **1**, wherein the target object is a fuselage, and wherein the fuselage is scanned so that the captured images of the slice extend from a midline of an underbelly of the fuselage to a midline of a top of the fuselage.

15. An inspection system, comprising:

an inspection cart, comprising:

a tower supporting a track;

a carriage movable along the track, the carriage supporting a 3D scanner having at least two cameras and a projector; and

optical fiducials arranged relative to the track so that the optical fiducials are in a field of view of the 3D scanner during scanning of a target object.

16. The inspection system of claim **15**, further comprising:

one or more processors and one or more non-transitory memory devices storing a program, which, when executed by any combination of the one or more processors, causes the one or more processors to perform an operation, the operation comprising:

causing the carriage to move along the track so that the 3D scanner is moved relative to the target object, causing, as the 3D scanner is translated, the 3D scanner to scan the target object to capture images of at least a slice of the target object;

stitching together the captured images of the slice of the target object to create a slice image, the captured images being stitched together according to position targets provided by the optical fiducials; and detecting one or more defects on the target object based on the slice image.

17. The inspection system of claim **16**, wherein the operation further comprises:

causing the projector of the 3D scanner to project one or more defect indicators onto the target object to indicate respective ones of the one or more defects detected on the target object.

18. The inspection system of claim **15**, wherein the track is a linear track and the optical fiducials include a first fiducial rail and a second fiducial rail that are spaced from one another and arranged parallel to the linear track.

19. The inspection system of claim **15**, wherein the target object is a fuselage, and wherein the track is a curved track that extends above and below the fuselage.

20. An inspection cart, comprising:

a tower supporting a track;

a carriage movable along the track, the carriage supporting a 3D scanner having at least two cameras and a projector; and

optical fiducials arranged relative to the track so that the optical fiducials are in a field of view of the 3D scanner during scanning of a target object.

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